

Appendix - AI Usage

AUTHOR

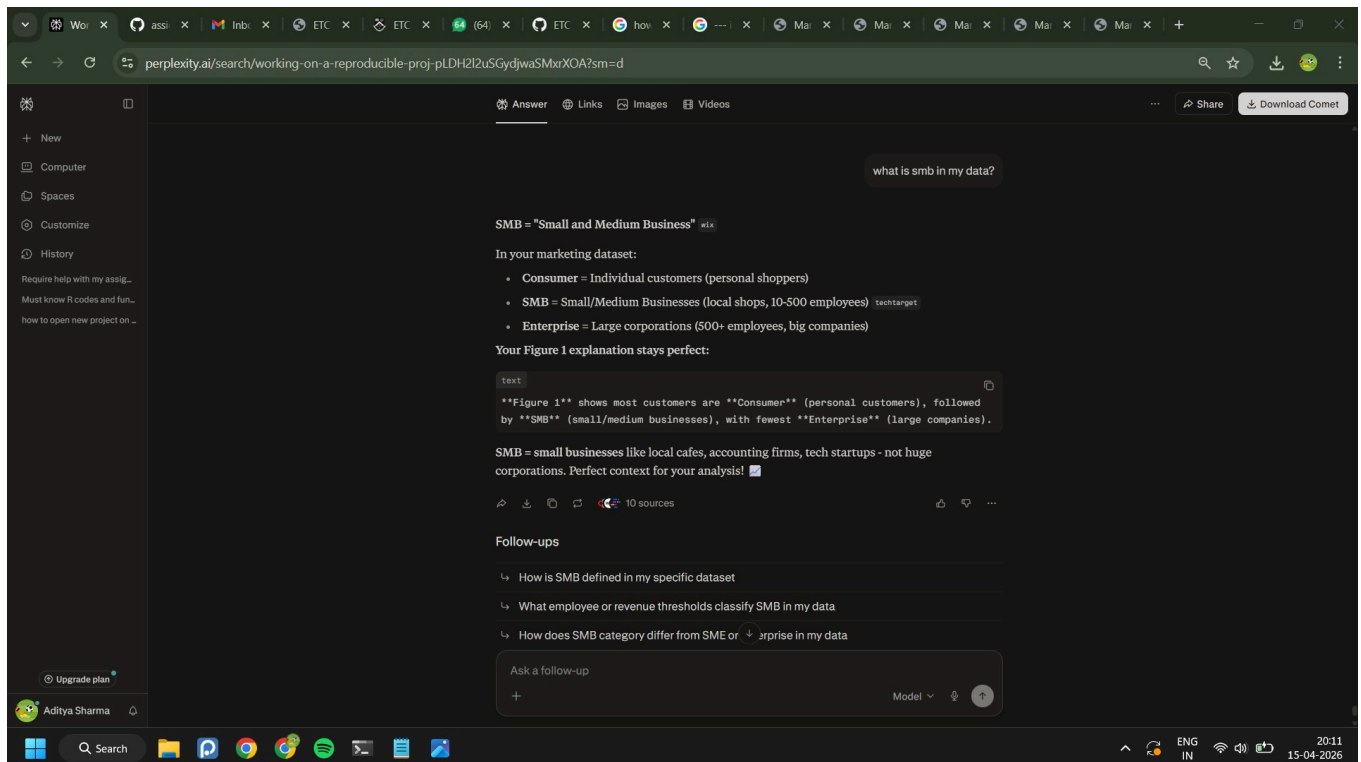
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AI Assistance Used

This analysis used **Perplexity AI** to:

1) Explain of Business terms

- **Customer Segments:** Explained SMB = Small/Medium Businesses



2) Fix grammatical/punctuation errors in my write-up

perplexity.ai/search/working-on-a-reproducible-proj-pLDH2I2uSGydwSMxXOA?sm=d

can you check for any grammatical and punctuation errors in the results section of my report? here is what i have typed:

RESULTS

Table 1 shows companies spent about \$142 per customer on marketing (median \$140), while customers bought around \$143 worth of products on average (median \$139). The revenue almost roughly equals the marketing cost here which suggests that there is a net profit (before company costs)

Figure 1 shows most customers are direct "Consumer" type (not businesses).

Figure 2 shows a line going "upwards". This means: "more marketing money = more customer purchases".

Conclusion: Higher marketing spend "does" lead to higher sales

Show less

Grammar & Punctuation Fixes

Current version has these issues:

- "almost roughly equals" = redundant
- "direct 'Consumer' type" = awkward phrasing

Ask a follow-up

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Corrected Results Section

RESULTS

Table 1 shows companies spent \$142 per customer on marketing (median \$140), while customers bought \$143 worth of products on average (median \$139). The revenue nearly equals marketing costs, suggesting the campaign "broke even" (before other expenses).

Figure 1 shows "most customers are Consumers" (individual buyers, not businesses).

Figure 2 shows an "upward trend line". "More marketing spend = higher customer purchases" (positive correlation).

Key fixes:

- Removed "almost roughly" → "nearly equals"
- "direct Consumer type" → "most customers are Consumers"
- Moved Conclusion to its own section
- Consistent bold formatting
- Professional phrasing throughout

Perfect now - clear, concise, academic!

Ask a follow-up

3) Code enhancement (to change bar graph colour)

I have written a code for a bar graph with 3 bars. I need enhancements to this code which shows all 3 bars in different colours for better visual aesthetics and comparison. here is my code

```
ggplot(cleaned_data, aes(x = segment, fill = segment)) +
  geom_bar() +
  labs(
    title = "Customer Segments",
    x = "Segment",
    y = "Count"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Reviewing the provided bar chart code to suggest color-enhanced variations for better comparison

```
ggplot(cleaned_data, aes(x = segment, fill = segment)) +
  geom_bar() +
  labs(
    title = "Customer Segments",
    x = "Segment",
    y = "Count"
  ) +
  scale_fill_manual(values = c("Consumer" = "#2E86AB",
                              "SMB" = "#A23872",
                              "Enterprise" = "#F18F01")) +
  theme_minimal() +
  theme(legend.position = "none")
```

Ask a follow-up

4) Help with the code to create an analysis table to compare Marketing Spend and Purchase amount

suggest how to create an analysis table to compare marketing spend and purchase amount by customer on R. it should have metrics such as mean and median. explain the sequence too

Libraries: dplyr, gt

Sequence (5 steps):

1. `library(dplyr) + library(gt)` - Load packages
2. `summarise(across(...))` - Calculate stats on both columns

```
cleaned_data %>%
  summarise(across(c(marketingspend, purchaseamount),
    list(Mean = ~mean(.x, na.rm=TRUE),
         Median = ~median(.x, na.rm=TRUE))))
```

3. `pivot_longer()` - Stack into Variable/Metric format
Clue: `names_sep = "_", names_to = c("Variable", "Metric")`
4. `pivot_wider()` - Spread Metrics as rows
Clue: `names_from = Variable, values_from = value`
5. `gt()` + `fmt_currency()` - Format as nice table

Pro tip: Add `round(.x, 1)` inside `list()` functions for 1 decimal place.

Ask a follow-up

5) To clear confusion regarding the sequence of usage of a function (is.na)

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which part of my code in a chunk does is filter(!is.na) appear in, after mutating and creating columns or before. show with a random example

Answer Links Images Videos

`filter(!is.na())` goes AFTER `mutate()`

Sequence: `mutate()` first → `filter()` second

Example:

```
cleaned_data <- raw_data %>%  
  # 1. FIRST: mutate() - CREATE new columns (may introduce NAs)  
  mutate(  
    segment = case_when(...), # Creates segment column  
    marketingspend = some_calculation # May create some NAs  
  ) %>%  
  # 2. SECOND: filter() - REMOVE the NAs you just created  
  filter(!is.na(segment), # Remove rows where segment is NA  
         !is.na(marketingspend)) # Remove rows where spend is NA
```

Why this order?

1. `mutate()` creates new columns with potential NAs
2. `filter()` cleans up those NAs after they're created

Your code structure:

```
mutate(segment = ..., marketingspend = ...) %>% # Step 1  
filter(!is.na(segment), !is.na(marketingspend)) # Step 2 ← HERE
```

Ask a follow-up

Model

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ENG IN 20:18 15-04-2026